

Ambuj Kumar Data Engineer

Texas | ambujkumar2511@gmail.com | +19453389264 | [LinkedIn](#)

Summary

Data Engineer with 6+ years of experience building scalable data platforms and AI-driven pipelines across Azure and AWS. Expert in Python, Spark, Databricks, Kafka, and Snowflake, delivering secure, governed data systems. Proven impact improving data reliability, analytics performance, and ML platform scalability while processing millions of records daily for enterprise and financial services.

Skills

Programming: Python, SQL, PySpark, T-SQL, Shell Scripting

Frameworks & Libraries: Apache Spark, Pandas, PyTorch, TensorFlow Lite, FastAPI, dbt, Great Expectations

Cloud & DevOps: Azure (Databricks, Data Factory, Synapse, ADLS Gen2, Blob Storage, Purview, Azure Functions, Azure DevOps, Azure Monitor, Azure OpenAI), AWS (S3, Lambda, Glue), Docker, Kubernetes, CI/CD, Jenkins, Git

Big Data & ETL: Apache Kafka, Spark Streaming, Databricks (Delta Lake, Unity Catalog), ETL/ELT Pipelines, Snowflake, Azure Synapse, SQL Server, PostgreSQL, Redis

Data Engineering: Batch & Streaming Data Pipelines, Data Ingestion Frameworks, REST API Integration, Medallion Architecture, Lakehouse Architecture, Dimensional Modeling (Star/Snowflake), Schema Evolution, Data Partitioning

AI & ML Data Systems: Feature Engineering Pipelines, RAG Pipelines, Vector Databases (FAISS, Redis), Model Inference APIs, Azure AI/OpenAI Services

Data Governance & Quality: dbt, Great Expectations, Azure Purview, Data Lineage, Metadata Management, Schema Validation, Data Observability, Row-Level Security

BI & Analytics: Power BI, Tableau, Looker, Looker Studio, Qlik Sense, QlikView, MicroStrategy, SAP BusinessObjects, IBM Cognos, Apache Superset, Executive KPI Dashboards, Data Modeling for BI

Orchestration & Workflow: Apache Airflow, Azure Data Factory Pipelines, Workflow Automation, Pipeline Monitoring, Alerting Systems

Collaboration: Agile Methodologies, Cross-Functional Collaboration, Data Platform Architecture, Stakeholder Reporting

Professional Experience

Abacus Group LLC — Data Engineer

May 2025 – Present | USA

- Architected enterprise data platform using Azure Databricks, Azure Data Factory, Python, SQL Server integrating Salesforce, ServiceNow with RBAC governance, monitoring improving analytics accessibility 45%
- Developed distributed ingestion frameworks using Python, REST APIs, Airflow, Azure Functions enabling integrations achieving 99.9% reliability with automated logging, alerting, pipeline monitoring
- Implemented enterprise data governance using dbt, Great Expectations, Azure Purview enabling validation, lineage, access control improving enterprise data trust and compliance 40%
- Built AI-driven data observability systems using Python, Databricks, Pandas, Azure OpenAI detecting anomalies early with monitoring dashboards reducing enterprise data incidents 35%
- Designed hybrid cloud architecture integrating SQL Server, Azure Blob Storage, Databricks, Azure Data Factory enabling scalable ingestion with secure data access improving efficiency 50%
- Optimized warehouse models and SQL workloads using Azure Synapse, SQL Server, Power BI with query auditing, monitoring improving analytics performance 45%

Bank of America — AI Data Engineer

Aug 2023 – Jul 2024 | USA

- Designed scalable GenAI pipelines using Python, Spark, Databricks processing 10M+ financial records daily implementing governance controls, monitoring and audit logging
- Built RAG-based retrieval systems using FAISS, Redis, PostgreSQL vector stores improving AI response accuracy 35% with secured access control and model monitoring
- Developed model inference APIs using FastAPI and Kubernetes reducing latency 40% with authentication, logging, and runtime monitoring ensuring secure AI deployments

- Implemented CI/CD pipelines using Git, Jenkins, Docker, Kubernetes, Azure DevOps reducing deployment time 50% with security checks and pipeline observability
- Optimized distributed Spark and SQL workloads improving runtimes 45% while implementing performance monitoring, logging, and infrastructure cost governance across clusters
- Mentored engineers delivering five enterprise AI platform features annually enforcing governance, security policies, monitoring improving platform reliability and compliance 35%

PwC — Cloud Data Engineer

May 2019 – Aug 2023 | India

- Architected enterprise data platforms using Python, Spark, Databricks, Snowflake building scalable ETL pipelines implementing governance, access control improving data accessibility 45%
- Developed ingestion frameworks using Python, REST APIs, Airflow, Kafka enabling integrations achieving 99.9% pipeline reliability with automated logging, monitoring, alerting systems
- Optimized distributed Spark and SQL workloads improving runtimes 40% implementing performance monitoring, pipeline observability and infrastructure cost governance
- Implemented enterprise data governance frameworks using dbt, Great Expectations and catalog tools enabling validation, lineage, compliance improving data quality 35%
- Designed streaming architectures using Kafka, Spark Streaming processing 20M+ events daily implementing secure pipelines, monitoring supporting fraud detection analytics
- Built cloud-native lakehouse using Azure Databricks, Azure Data Factory, ADLS enabling scalable analytics platforms improving performance 50% with governance controls

HCL Technologies — Data Engineer Intern

Jun 2018 – Sep 2018 | India

- Implemented Edge AI filtering pipelines using Python, TensorFlow Lite, and Azure IoT Edge reducing cloud data transit costs 60% while maintaining real-time processing
- Developed Python-based data preprocessing pipelines using Pandas and NumPy cleaning high-volume edge sensor datasets improving data quality and processing efficiency 40%
- Built scalable ETL pipelines using Python, Apache Kafka, and Azure Data Factory streaming filtered device telemetry into cloud storage processing 5M+ records daily
- Monitored edge-to-cloud data pipelines using Python, Spark, and Azure Monitor validating data integrity improving pipeline reliability 35% for real-time analytics systems

Education

The University of Texas at Dallas	Aug 2024 – Dec 2025
Master of Information Technology & Management (MIS) – Data Engineering Track	
SRM University	May 2015 – Jun 2019
Bachelor of Technology – Computer Science Engineering	